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6.2.2 Error in storing a real number as a floating point number

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Let's look at what kind of error is introduced when we store a real number as a floating-point number. So let's assume that the number we start with is a positive real number. And that number can be

5:36 / 5:36

2.0x

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Homework 6.2.2.1

1/1 point (graded)
Assume a floating point number system with $\beta = 2$, a mantissa with t digits, and truncation when storing.

- Write the number **1** as a floating point number in this system.
- What is the ϵ_{mach} for this system?

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Solution:

For a complete solution, see [the notes](#).

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